

Handout: Ternary Relationships with Cardinality and Participation Constraints

Introduction

In the Chen ER model, a **ternary relationship** connects three entities. Unlike binary relationships, ternary relationships require special attention to:

- Cardinality constraints (1 vs many)
- Total vs partial participation

This handout uses the example:

`Prescribes(Doctor, Patient, Medication)`

Cardinality Constraints in Ternary Relationship

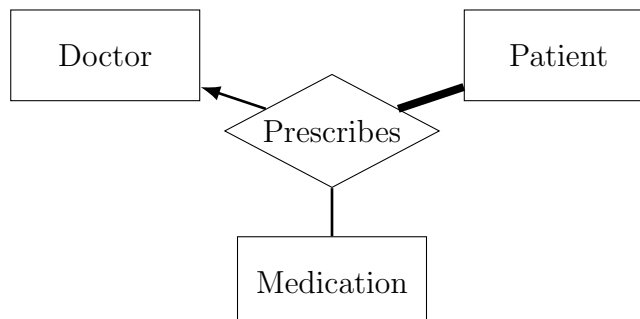
In ternary relationships, cardinality is expressed as:

For each combination of the other two entities, how many instances of this entity can participate?

For example:

- For each (Patient, Medication), how many Doctors? $\rightarrow 1$
- For each (Doctor, Medication), how many Patients? $\rightarrow \text{many}$
- For each (Doctor, Patient), how many Medications? $\rightarrow \text{many}$

Diagram: Cardinality and Participation



Meaning of the Diagram

- $\rightarrow$ **to Doctor**: For each (Patient, Medication) pair, only **one Doctor** can prescribe.
- --- **to Patient**: For each (Doctor, Medication) pair, **many Patients** can receive prescriptions.
- --- **to Medication**: For each (Doctor, Patient) pair, **many Medications** may be prescribed.

Participation Constraints

- ****Total participation**** is shown with a **double line**.
- ****Partial participation**** (default) uses a regular line.
- Total participation means *every entity instance must participate in at least one relationship*.

Summary

- Use arrows ($\rightarrow$) to denote **1 cardinality**, and lines (---) for **many**.
- Always interpret ternary relationship cardinality by **fixing the other two entities**.
- Use **double lines** for total participation.